

# Humanoid Robots in Healthcare: Current Systems, Research Frontiers, and Clinical Opportunities

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Humanoid robots capable of dexterous bimanual manipulation and natural language instruction are commercially available, yet no full-size bipedal humanoid holds FDA clearance or CE marking for any clinical application. This survey covers the 2022–2026 humanoid healthcare literature: 14 qualifying papers identified, 25 platforms assessed, regulatory pathways analyzed, and ISO 16290:2013 TRL criteria assigned to five clinical domains. All domains are at TRL 3–4. Three roles—teleoperated clinical assistant, rehabilitation demonstrator, and eldercare companion—have the strongest path to TRL 5. The binding barrier is the lack of accepted bipedal gait safety standards for patient-proximate environments.

CCS Concepts: • **Computer systems organization** → **Robotics**; • **Computing methodologies** → *Artificial intelligence*; • **Social and professional topics** → *Medical informatics*.

Additional Key Words and Phrases: humanoid robots, healthcare robotics, embodied AI, large language models, human-robot interaction, clinical deployment, medical robots, technology readiness level, regulatory pathway

## ACM Reference Format:

Anonymous Author(s). 2026. Humanoid Robots in Healthcare: Current Systems, Research Frontiers, and Clinical Opportunities. *ACM Comput. Surv.* 0, 0, Article 0 (2026), 30 pages.

## 1 Introduction

The most capable humanoid robots ever built are commercially available today. Figure 02, Boston Dynamics Atlas (electric), Tesla Optimus Gen 2, Unitree G1, Fourier GR-2, and more than a dozen comparable platforms combine bipedal locomotion, dexterous multi-fingered manipulation, and onboard sensing into systems that operate in spaces designed for humans, handle objects designed for human hands, and respond to instructions in natural language. Twelve of thirteen major commercial humanoid platforms have confirmed or announced large language model integration into their control architecture. In the laboratory, a Unitree G1 teleoperated by a non-clinician completed seven distinct clinical procedures—auscultation, ultrasound, IV catheter insertion, and four others—in a controlled experimental setting [4]; a second system operated as a surgical first assistant in a cadaveric bilateral sphenoidectomy [16]. Yet not one of these systems holds FDA clearance or CE marking for any clinical application. No registered Phase I, II, or III clinical trial has enrolled a patient for a study involving a full-size bipedal humanoid platform. The technical capability exists at a level that warrants serious clinical investigation. The clinical deployment does not. This review examines why.

The disparity between demonstrated capability and clinical deployment is not a permanent condition; it is a developmental phase. Two technological developments converged after 2022 to move the humanoid healthcare question from speculative to tractable. The first is hardware: commercial humanoid platforms crossed the payload, dexterity, and locomotion threshold required to attempt clinical and care tasks in their natural environments. Actuator and sensor configurations now exist that allow manipulation of clinical instruments—stethoscopes, ultrasound probes, laparoscopes—and navigation of clinical spaces at human working height and stride. The second is software: the integration of large

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2026. Manuscript submitted to ACM

language models and vision-language-action (VLA) models removed the per-task explicit programming requirement that previously made general-purpose humanoid deployment impractical. Instruction-following, task generalization, and adaptive response to unstructured environments became achievable without the months-long engineering cycles that earlier control architectures demanded. Both were necessary; neither alone was sufficient. The result is a field that moved, in roughly three years, from proof-of-concept locomotion demonstrations to laboratory execution of clinical procedure sequences. The literature reviewed here covers that period: 2022 to early 2026.

The existing review literature does not yet address this convergence. The closest prior work is Cunha et al. [17], a 2025 scoping review of 27 studies on care provided by humanoid robots, with a search cutoff of November 2023—predating the LLM/VLA integration that defines the current period entirely. Earlier reviews of humanoid robots in healthcare [5, 40] and service robots for healthcare delivery [21, 38, 46] treat humanoid and non-humanoid platforms interchangeably, obscuring the form-factor-specific barriers that determine clinical deployability. Embodied AI reviews [32, 33, 37] cover the healthcare vertical as one application among many, without the platform-specific and regulatory specificity that the clinical deployment question requires. This review addresses the gap directly: it is the first synthesis of the 2022–2026 humanoid-specific clinical and care literature that integrates a platform capability assessment, a regulatory pathway analysis grounded in current FDA and EU MDR frameworks, and a per-domain technology readiness evaluation using ISO 16290:2013 TRL criteria—with clinical deployment as the organizing question throughout.

The scope of this review is defined by platform type. Included are bipedal, free-standing humanoid robots of approximately 150–200 cm height, designed for general-purpose manipulation and locomotion in human-scale environments, with literature coverage from 2022 to March 2026. Several categories are excluded with specific rationale. Wheeled social robots—including Pepper (SoftBank Robotics) and ARI (PAL Robotics)—are excluded because their mobility constraints and form-factor dynamics differ from bipedal platforms in ways relevant to clinical deployment analysis; where a wheeled platform appears in a reviewed paper, it is retained with an explicit scope caveat. Sub-100 cm research platforms including NAO are excluded for the same operational envelope reason. Surgical robotic arms (da Vinci) are excluded as task-specific fixed-base systems with entirely separate regulatory histories. Powered exoskeletons are excluded as patient-worn assistive devices operating under a different deployment model. RHP Friends [6], a purpose-built bipedal nursing humanoid, is the only non-commercial academic platform retained without caveat; it meets all inclusion criteria.

The central contribution of this review is a regulatory-grounded readiness analysis, not a literature catalog. The key idea is that clinical deployment barriers differ by role: the HCDR (Humanoid Clinical Deployment Readiness) framework introduced here classifies clinical roles by autonomy level, patient-contact intensity, and regulatory complexity, and establishes that the binding constraint is role-specific—a finding that changes how investment priorities should be sequenced. Supporting this analysis are: a systematic survey of 14 qualifying peer-reviewed papers (arXiv, PubMed, IEEE Xplore, ACM Digital Library, and Google Scholar, 2022–early 2026)—the most complete such survey to date; a structured capability assessment of 13 commercial and 12 academic platforms with deployment status verified against primary sources, including the finding that no platform holds clinical regulatory clearance; a per-domain TRL assessment applying ISO 16290:2013 to five application domains, identifying the specific engineering threshold required for each; and a regulatory pathway analysis of FDA, EU MDR, and ISO frameworks that names the jurisdictional gaps that must be resolved before first regulatory clearance can occur.

The paper is organized as follows. Section 2 describes the systematic search and selection protocol. Section 3 provides background on the healthcare imperative, the humanoid form-factor rationale, and the LLM/VLA inflection point. Section 4 surveys commercial and academic humanoid platforms. Section 5 reviews the clinical and care research literature by application domain. Section 6 analyzes the technical and translational barriers. Section 7 synthesizes a

readiness assessment, proposes a clinical deployment readiness framework, identifies near-term deployable roles, and outlines practitioner guidance and the regulatory roadmap. Section 8 concludes.

## 2 Survey Methodology

**Search strategy.** Searches were conducted across five databases: arXiv (cs.RO, cs.AI, eess.SY), PubMed, IEEE Xplore, ACM Digital Library, and Google Scholar. The primary search string combined platform and application terms: (*humanoid robot OR bipedal robot*) AND (*healthcare OR medical OR clinical OR nursing OR rehabilitation OR surgery*). Domain-specific secondary searches were run for each clinical application area (Section 5). Coverage: January 2022 through March 2026. Citation-list hand-search of all included papers was conducted to capture works not returned by the primary query.

**Inclusion criteria.** A paper qualified if it: (1) involved a bipedal or human-form humanoid robot meeting the scope definition in Section 1; (2) reported empirical results in a clinical, care, or clinical-simulated context; and (3) was peer-reviewed or a major-venue preprint with evidence of submission.

**Exclusion criteria.** Papers were excluded if they used only wheeled platforms (Pepper, ARI), sub-100 cm platforms (NAO), or non-humanoid systems without bipedal evidence; reported simulation-only evaluation; presented hardware or controls contributions with no clinical context; or were prior surveys (catalogued separately in the novelty analysis).

**Screening and yield.** Title and abstract screening of 147 candidate papers yielded 52 full-text reviews. Full-text application of inclusion and exclusion criteria produced 14 qualifying papers. Primary exclusion reasons: non-qualifying platform (89 papers eliminated at title/abstract stage), simulation only (27), no clinical task context (17). The sparsity of the qualifying set is itself a primary finding: the evidence base for bipedal humanoid clinical research is narrow and concentrated.

**Platform survey.** Concurrent with the literature search, a structured capability survey identified 13 commercial and 12 academic humanoid platforms meeting scope criteria. Specifications were verified against primary manufacturer documentation or peer-reviewed publications; undisclosed values are reported as N/A.

**Regulatory analysis.** The regulatory pathway analysis draws on primary source documents: FDA De Novo guidance (21 CFR 513(f)(2)), EU Medical Device Regulation (2017/745), EU AI Act (2024/1689), ISO 13482:2014, and ISO/TS 15066:2016.

## 3 Background

Before assessing the clinical evidence, three core developments must be understood: the healthcare workforce crisis that creates demand for robotic assistance, the humanoid form-factor argument that makes bipedal platforms specifically relevant, and the LLM/VLA inflection that makes general-purpose clinical deployment conceivable for the first time. Two supplementary elements complete the context: a historical trajectory situating current platforms within the arc of bipedal robotics, and a structured comparison of humanoid against non-humanoid robotic alternatives.

### 3.1 The Healthcare Imperative

The global healthcare workforce is under structural strain that technology alone cannot resolve, but that technology may help mitigate. The World Health Organization projects a shortage of approximately 10 million healthcare workers by 2030, concentrated primarily in low- and middle-income countries but with significant pressure also on high-income systems where demographic aging compounds absolute workforce insufficiency [65]. The root cause is well understood: populations in Organisation for Economic Co-operation and Development (OECD) countries are aging faster than

working-age cohorts are growing [45]. In Japan—the most extreme case—persons aged 65 and over already represent approximately 30% of the total population, a proportion that continues to rise under below-replacement fertility rates [11]. The METI 2023 robotics roadmap explicitly identifies nursing-capable robots as a national strategic target by 2030, reflecting governmental acknowledgment that the eldercare workforce gap is structurally unsolvable by recruitment alone.

Nursing workforce shortages present a compounding challenge [29]. The care gap operates at two intersecting levels.

First, the absolute number of care hours demanded rises with the prevalence of chronic disease, multimorbidity, and frailty in an older population: an individual with dementia and comorbid heart failure requires qualitatively more nursing time than a healthy adult, and the number of such individuals is growing rapidly. Second, the tasks that constitute professional care—physical repositioning of patients, vital sign monitoring, medication administration, rehabilitation exercise guidance, and social engagement—are labor-intensive and resist substitution by currently deployed information technology. Electronic health record systems and remote monitoring platforms reduce administrative overhead but do not reduce the need for physical presence at the bedside.

Automation is increasingly discussed as a necessary complement—not a replacement—for human care. The framing matters: the goal is not to displace nurses or physicians, but to extend their effective reach by delegating tasks that are physically demanding, repetitive, or logistically routine, freeing trained clinicians for higher-order judgment and human connection. Wheeled mobile robots already handle medication transport and specimen delivery in hundreds of hospitals. Exoskeletons assist rehabilitation and reduce caregiver musculoskeletal strain. Social robots provide cognitive stimulation and companionship in dementia care settings. Each of these automation layers is real and documented. What remains largely unrealized is a robot capable of performing the physical care tasks—patient repositioning, standing assist, equipment handling, clinical examination—that currently require a human body in proximity to the patient. This is where the humanoid form factor becomes relevant.

### 3.2 Humanoid Form Factor: Rationale for Healthcare

For the purposes of this survey, a *humanoid robot* is defined as a bipedal, free-standing robot with a human-proportioned morphology: head, torso, two arms, and two legs, typically ranging from 150 to 200 cm in height. This definition explicitly excludes social robots such as Pepper (120 cm, wheeled base) and NAO (58 cm, child-scale), surgical manipulator arms such as the da Vinci system, and wearable exoskeletons. The rationale for this scope restriction is not hierarchical—social robots and surgical robots are clinically important and are discussed in Section 3.5—but conceptual: the form-factor hypothesis is specifically about whether human-scale bipedal morphology confers advantages in human-built environments that smaller or non-mobile systems cannot provide.

Three distinct lines of argument support the humanoid form factor’s suitability for healthcare deployment.

**Environmental compatibility.** Hospitals, care homes, and residences are built for human-sized bipedal operators. Doorways, corridors, stairways, beds, examination tables, supply cabinets, and clinical instruments are all dimensioned for a person approximately 160–180 cm tall with two-handed reach. A bipedal humanoid can, in principle, navigate these environments without infrastructure modification: walk through standard doorways, climb stairs, reach across a standard hospital bed, and operate existing clinical tools (stethoscope, bag-valve mask, otoscope, ultrasound probe) designed for human hands. The technical demonstration by Atar et al. [4] explicitly grounds this observation: the Unitree G1 demonstrated seven medical procedures using existing clinical instruments, without facility modification.

**Neuroscientific basis for human-robot interaction.** Action observation therapy (AOT)—a rehabilitation technique in which patients observe biological motion to activate sensorimotor circuits—rests on the mirror neuron hypothesis:

observing purposive action activates similar neural circuits as performing that action [51]. Nguyen et al. [41] provide the most direct empirical evidence available: EEG recordings from healthy participants during observation of a humanoid robot performing reaching and grasping actions showed sensorimotor response patterns comparable to those elicited by observing a human agent perform the same actions. This finding provides neurophysiological grounding for the hypothesis that the humanoid form specifically may be required for AOT efficacy. A complementary behavioral finding comes from Sayis and Gunes [54], who demonstrated that only the humanoid robot condition—not a voice-only assistant—produced improvements in mood and increased self-disclosure in a therapeutic writing study ( $n = 42$ ), attributing the difference to the physical embodiment of the humanoid.

**User acceptance and eye-level interaction.** Patients in clinical settings are frequently in positions of physical vulnerability. The SPRING project [2] (*scope caveat*: ARI is a wheeled platform, not bipedal) found that users in a real geriatric day-care facility explicitly cited upright, eye-level conversational presence as important for dignified interaction—a form-factor finding directly relevant to bipedal humanoid design regardless of the mobility substrate. Additional contextual evidence comes from research on the HRP-4C platform, which demonstrated that gait naturalness and facial expression generation systematically affected observer ratings of approachability and social presence [14].

These findings collectively support—but do not yet conclusively prove—the form-factor hypothesis at clinical scale. The evidence is encouraging; it is not sufficient to substitute for clinical validation.

### 3.3 The LLM/VLM/VLA Inflection Point

Prior to approximately 2022, humanoid robots required explicit, hand-crafted programming for every behavioral task. A robot capable of performing a stethoscope placement procedure had to be programmed, step by step, for that specific procedure. Deploying the same robot for a second procedure required a substantially new programming effort. This approach fundamentally limited humanoid deployment in clinical environments, which are intrinsically unstructured.

The emergence of large language models (LLMs), vision-language models (VLMs), and vision-language-action (VLA) models has changed this constraint. GPT-4 and its successors demonstrated that natural language can serve as a general-purpose task specification interface, capable of interpreting novel instructions and decomposing them into action plans [44]. Physical Intelligence’s  $\pi_0$  and  $\pi_{0.5}$  models introduced flow-based VLA architectures mapping language instructions and visual observations directly to robot actions, trained on thousands of hours of real robot demonstrations [7]. OpenVLA [27] demonstrated that open-source, community-accessible VLA models could achieve meaningful generalization from limited task demonstrations. RT-2 [9] established that vision-language foundation models trained on internet-scale data transfer task-relevant knowledge to robotic control without task-specific fine-tuning. Embodied AI surveys covering the healthcare vertical confirm that this software inflection is accelerating clinical relevance [32, 33, 37].

Yuan et al. [66] demonstrated this integration pattern in a dementia care context, combining reinforcement learning with a large language model to generate adaptive, personalized robot behavior. Although Pepper is a wheeled robot (not within our bipedal scope), the LLM integration framework—specifically its ability to model patient-specific cognitive-emotional states and adapt behavior accordingly—is directly applicable to bipedal humanoid platforms.

The commercial humanoid wave of 2022–2026 coincides with this LLM inflection, and the coincidence is not accidental. Twelve of the thirteen commercial humanoid systems surveyed in Section 4 have confirmed or announced LLM/VLA integration as of 2026. The design assumption of current commercial humanoids is language-conditioned, generalist behavior: these systems are not being built as specialized single-task machines, a prerequisite for clinical task diversity.

### 3.4 Historical Evolution of Humanoid Robotics

The history of humanoid robotics over the past three decades can be organized into three eras, followed by the nascent healthcare entry period this review covers.

**Locomotion proof-of-concept era (1996–2010).** Honda’s ASIMO (2000) demonstrated that dynamic bipedal walking was achievable with then-current actuation and control technology [14, 53]. The AIST HRP series (HRP-2 through HRP-5P, 1998–2019) established the research infrastructure for whole-body motion planning [26]. Preview control of the zero-moment point (ZMP) provided the mathematical foundation for stable bipedal gait generation [25]. These systems proved that bipedal locomotion was a solvable engineering problem, with no meaningful manipulation capability or path to clinical relevance. ASIMO was retired in 2022.

**Manipulation and integration era (2010–2020).** The DARPA Robotics Challenge (DRC, 2012–2015) revealed the gap between locomotion capability and reliable task execution. Boston Dynamics’ hydraulic Atlas became the reference platform for whole-body manipulation research and produced dynamic capability demonstrations while revealing that force-controlled manipulation at clinical precision remained largely unsolved [14]. The decade consolidated learning-based locomotion control, particularly deep reinforcement learning policies trained in simulation and transferred to hardware. Recent work demonstrates that transformer-based policies trained on motion-capture data enable zero-shot locomotion on commercial humanoids [49].

**Commercial scale-up era (2022–present).** The confluence of electric actuator maturation, learning-based locomotion policies, and LLM integration produced the current commercial humanoid wave. Thirteen distinct commercial humanoid systems reached hardware maturity between 2022 and 2026. Global investment in humanoid robotics grew rapidly through 2024, driven by the convergence of electric actuator maturation and LLM integration, with confirmed commercial deployments concentrated in automotive and logistics applications. As of March 2026, none of these thirteen systems has been deployed in a confirmed clinical patient care context.

**Healthcare entry point (2024–2026).** Atar et al. [4] demonstrated bimanual teleoperation of a Unitree G1 across seven clinical procedures with approximately 70% aggregate success—the first systematic evaluation of a current-generation commercial humanoid against a clinically meaningful task battery. Cho et al. [16] extended this to a cadaveric endoscopic sinus surgery at Johns Hopkins University—the first documented use of a modern commercial bipedal humanoid in an actual (cadaveric) surgical procedure. This survey covers this nascent period.

Figure 1 situates these milestones in the broader thirty-year arc of humanoid robot development.

### 3.5 Humanoid Robots vs. Non-Humanoid Alternatives in Healthcare

A principled comparison with existing clinical robot categories is necessary to locate the humanoid value proposition precisely.

**3.5.1 Social Robots: NAO, Pepper, PARO.** Social robots have the deepest clinical evidence base in robotics. NAO (SoftBank/Aldebaran, 58 cm, 25 DOF) has accumulated over 1,000 peer-reviewed publications, with clinical use spanning autism spectrum disorder therapy, elderly care, and hospital patient reception [10]. Pepper has been deployed in approximately 2,000 Japanese healthcare facilities for dementia cognitive stimulation and companion interaction, with embodied conversational agent frameworks increasingly integrated for clinical applications [17, 48, 59]. The PARO therapeutic seal robot has demonstrated psychosocial benefits in randomized controlled trials with elderly residents [64]. These platforms set the deployment high-water mark for robots in clinical care settings [1]. However, their form-factor

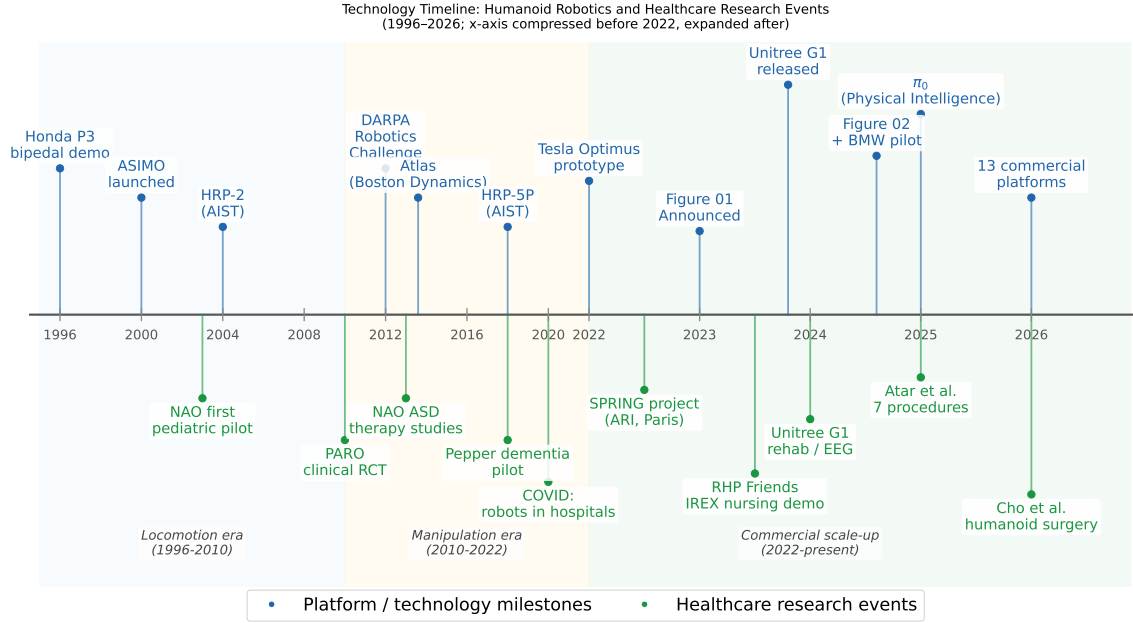


Fig. 1. Technology timeline: humanoid robotics platform milestones (upper track, blue) and healthcare research events (lower track, green), 1996–2026. Shaded regions demarcate three developmental eras: locomotion proof-of-concept (1996–2010), manipulation and integration (2010–2022), and commercial scale-up (2022–present). Healthcare research events lag platform capability by approximately two years in each era.

constraints are real: NAO at 58 cm cannot reach adult-height work surfaces or perform physical care tasks for adults; Pepper on a wheeled base cannot navigate stairways or use standard clinical beds.

**3.5.2 Surgical Robots: da Vinci.** The da Vinci Surgical System is FDA-cleared (Class II, 510(k)) with over 10 million procedures completed across urology, gynecology, and general surgery [28, 57]. Robotic surgery adoption has grown substantially over the past decade, though adverse event analysis reveals the importance of systematic safety monitoring even for established platforms [3]. The historical trajectory from laboratory prototype to clinical instrument provides context for humanoid deployment [13]. Surgical robots are not competitors to humanoid clinical assistants; they are complementary systems addressing a fundamentally different need [38]. Da Vinci is stationary, surgeon-controlled, and single-purpose. A humanoid clinical assistant operating in a post-surgical ward addresses the nursing and logistics tasks that da Vinci cannot.

**3.5.3 Exoskeletons: ReWalk, HAL, Ekso Bionics.** Wearable robotic exoskeletons are FDA-cleared (Class II) for spinal cord injury rehabilitation, with Cochrane-level evidence supporting electromechanical-assisted training for walking recovery after stroke [36]. These devices are established tools in rehabilitation [59]. Exoskeletons assist the patient’s own movement and are not substitutable with a humanoid assistant: they are worn by the patient, require near-ambulatory capacity, and cannot lift or reposition a non-ambulatory patient. The GentleHumanoid framework [34] demonstrates the humanoid’s complementary role: learning upper-body compliance for sit-to-stand assistance addresses patients for whom exoskeletons are not an option.



Table 1. Humanoid Platform Capability Comparison (March 2026). N/A = not publicly disclosed. <sup>†</sup>Research prototype only, not clinical deployment. <sup>\*</sup>Not bipedal under this paper’s scope definition; included as deployment benchmark.

| Robot               | Type       | Ht      | Total DOF | Payload  | LLM Integration         | Open SW | Medical Deployment                   |
|---------------------|------------|---------|-----------|----------|-------------------------|---------|--------------------------------------|
| Figure 02           | Commercial | 168     | 28        | 20–25 kg | Yes (in-house VLA)      | No      | None                                 |
| Tesla Optimus G2    | Commercial | 173     | 28+       | 20 kg    | Yes (FSD-derived)       | No      | None                                 |
| BD Atlas (elec.)    | Commercial | ~165    | 28        | 30 kg    | Yes (Gemini, announced) | No      | None                                 |
| Agility Digit v4    | Commercial | 175     | N/A       | 16 kg    | Announced               | Partial | None                                 |
| Unitree G1          | Commercial | 132     | 23–43     | N/A      | Yes (VLA, open)         | Yes     | Research proto. [4, 16] <sup>†</sup> |
| Unitree H1          | Commercial | 180     | 19        | N/A      | Yes (open)              | Yes     | None                                 |
| 1X NEO Beta         | Commercial | 165     | 75        | 25 kg    | Yes (proprietary)       | No      | None                                 |
| Apptronik Apollo    | Commercial | 173     | N/A       | 25 kg    | Yes (Google AI)         | No      | None                                 |
| Sanctuary Ph. G8    | Commercial | 170     | N/A       | 25 kg    | Yes (Carbon/Microsoft)  | No      | None                                 |
| Fourier GR-2        | Commercial | 175     | 53        | N/A      | Announced               | Partial | None                                 |
| UBTECH Walker X     | Commercial | 130–145 | 41        | 10 kg    | Yes (multimodal LLM)    | No      | None                                 |
| AgiBot A2           | Commercial | 175     | 49+       | N/A      | Yes (proprietary)       | No      | None                                 |
| NEURA 4NE-1         | Commercial | ~175    | N/A       | 15–20 kg | Yes (cognitive AI)      | No      | None                                 |
| RHP Friends         | Academic   | 168     | 30        | N/A      | No                      | Partial | IREX 2023 nursing demo [6]           |
| NAO <sup>*</sup>    | Academic   | 58      | 25        | N/A      | No                      | Partial | HRI/ASD trials (social only)         |
| Pepper <sup>*</sup> | Academic   | 120     | N/A       | N/A      | No                      | Partial | 2,000+ Japan care facilities         |

**3.5.4 Wheeled Mobile Robots: TUG, Moxi.** Autonomous wheeled mobile robots address hospital logistics with confirmed operational deployments at major US health systems [21, 38]. These systems are purpose-built for point-to-point transport on flat pre-mapped surfaces. They cannot navigate stairs, interact with clinical equipment not specifically engineered for their interfaces, or perform physical patient care tasks. The humanoid argument relative to wheeled logistics robots is a division of labor argument: wheeled robots solve the logistics layer effectively; humanoids are needed for the physical assistance and care layer.

### 3.6 Section Summary

The four alternative categories collectively frame a finding that motivates this review: no existing clinical robot category combines (a) general ambulatory mobility in human-built environments, (b) bimanual dexterous manipulation of existing clinical tools, and (c) natural language-mediated task specification enabling adaptive behavior in unstructured settings. This combination is the humanoid value proposition for healthcare. The evidence reviewed in this section supports the form-factor hypothesis without constituting clinical proof. The remainder of this survey characterizes where the field stands, what has been demonstrated, and what must be resolved before that promise can be realized.

## 4 Current Humanoid Systems

The global humanoid robot field in 2024–2026 comprises two largely distinct populations: commercial platforms designed for industrial labor and a set of academic research systems whose provenance spans two decades of bipedal locomotion research. The central empirical finding of this section is that no confirmed clinical deployment of any full-size bipedal humanoid robot exists as of March 2026 [4, 16, 17].

Table 1 provides the full specification comparison for all thirteen commercial and selected academic platforms discussed in this section. The analysis below synthesizes capability patterns and clinical relevance across the field rather than characterizing individual platforms.



#### 4.1 Manipulation Maturity

Dexterous manipulation is the primary functional gap between current humanoid capability and clinical task requirements. Across the thirteen commercial platforms surveyed, total articulation ranges from approximately 3 DOF per arm (Agility Digit v4, designed for logistics bin-picking) to 75 total DOF (1X NEO Beta, with 22 DOF per hand). This spread reflects a fundamental architectural divergence: logistics-oriented platforms optimize for payload and cycle time, whereas platforms targeting service or research applications invest in hand complexity.

Payload capacity spans 10–30 kg across the commercial tier, well above the requirements for most clinical manipulation tasks (stethoscope, ultrasound probe, catheter, laparoscopic instrument). The binding constraint is not force output but force *control*: clinical contact requires sub-Newton precision at the contact surface, a requirement that no platform has validated against patient-contact standards such as ISO/TS 15066. The single exception is Sanctuary Phoenix Gen 8, which specifies 5 mN tactile force sensitivity—the only commercial platform with a disclosed haptic sensing specification relevant to clinical touch discrimination.

Documented medical procedure performance exists for one platform only: the Unitree G1, in two independent studies achieving approximately 70% aggregate task success across seven clinical procedures under teleoperation [4], and demonstrating endoscope holding in a cadaveric surgical context [16]. Both results are preclinical prototypes. The cross-platform inference is sobering: the most dexterous commercial platforms (NEO Beta, Sanctuary Phoenix) have no clinical task evidence, while the only platform with clinical evidence (Unitree G1) was selected not for its dexterity but for its open SDK and research accessibility. Manipulation capability is not the bottleneck; validated manipulation *for clinical contact* is.

#### 4.2 Locomotion and Clinical Environment Compatibility

The 2022–2024 generation of commercial humanoids completed a decisive architectural transition from hydraulic to electric actuation. All thirteen commercial platforms surveyed are fully electric, with brushless DC or high-torque servo actuators. This transition has three clinical implications: (1) reduced acoustic output enabling patient-proximate operation, (2) elimination of fluid-system leak risk in sterile environments, and (3) simplified field maintenance without hydraulic servicing. 1X NEO Beta specifies 22 dB operating noise—below the 30–35 dB threshold typical of hospital wards at night—making it the only platform with a disclosed acoustic specification relevant to patient care settings.

However, the clinical environment poses locomotion requirements that industrial deployments have not exercised. Hospital flooring transitions between vinyl, linoleum, ceramic tile, and threshold plates; patient room layouts require operation in spaces as narrow as 90 cm; and patient contact during transfer assistance imposes dynamic disturbance loads not present in logistics contexts. No commercial platform has published locomotion performance data in hospital environments, and no platform has been evaluated against IEC 62133 or IEC 60601 electrical safety standards in clinical settings.

Platform scale is a further structural constraint. The scope of this review sets a 150 cm lower bound for human-adult-care tasks; of the 13 commercial platforms, UBTECH Walker X (130–145 cm) and Unitree G1 (132 cm) fall at or below this threshold. The Unitree G1 is retained as the only platform with clinical evidence despite this limitation. Academic platforms designed for healthcare—including RHP Friends (168 cm, 54 kg, 30 DOF)—demonstrate that human-scale form factors can be achieved with explicit clinical design intent [6].

### 4.3 LLM/VLA Integration and Task Architecture

Twelve of the thirteen commercial platforms surveyed include announced or deployed LLM/VLA integration [14]. Architectures span three patterns: (1) in-house VLA foundation models trained on proprietary demonstration data (Figure 02, Tesla Optimus); (2) integration of open third-party models such as Google Gemini Robotics (Boston Dynamics Atlas), open VLA frameworks (Unitree G1 with LeRobot and OpenVLA support), or Microsoft-partnered AI (Sanctuary Phoenix via Carbon); and (3) multimodal LLM integration for instruction following without explicit VLA policy training (UBTECH Walker X).

This architectural diversity obscures a uniform empirical gap: no VLA or LLM system deployed on any humanoid platform has been trained on, evaluated against, or validated for clinical task performance. The largest open VLA training corpus (Open X-Embodiment, approximately 970,000 demonstrations) consists entirely of household tabletop manipulation and pick-and-place tasks. Medical procedures require physically and procedurally distinct demonstration data—instrument grasps under sterile technique, force profiles during tissue contact, workflow sequencing under infection control constraints—none of which appear in any published training corpus.

The implication for clinical deployment is concrete: LLM/VLA integration as currently implemented provides natural-language task specification and general manipulation priors, but does not constitute a path to autonomous clinical task execution. The Unitree G1’s teleoperation results [4, 16] are instructive precisely because they bypass VLA autonomy entirely: the cognitive and dexterous work is performed by a remote human operator, with the robot serving as a physical proxy. This teleoperation-first architecture may represent the viable near-term clinical deployment model for all platforms, regardless of their onboard AI sophistication.

### 4.4 Platform Accessibility for Clinical Research

Research access to humanoid platforms is heavily asymmetric. Unitree Robotics is the only major commercial manufacturer to release and maintain a fully open hardware SDK, which directly explains why the Unitree G1 is the only commercial platform with peer-reviewed clinical task evidence. All other commercial platforms operate under proprietary software stacks with no published API access for external researchers. Academic platforms—RHP Friends (30 DOF, purpose-built for nursing), iCub3 (54 DOF, full-body tactile skin, 40+ units in research labs)—provide broader research access but lack the scale and industrial backing needed for clinical deployment pathways.

The open-source software ecosystem relevant to humanoid clinical research is mature in general manipulation infrastructure but entirely absent in clinical-specific components. A systematic survey of prominent repositories yields four layers:

**Foundation models and VLA frameworks.** LeRobot [12] (HuggingFace; 22,179 stars; Apache 2.0) provides a hardware-agnostic imitation learning and VLA pipeline with direct Unitree G1 support. Physical Intelligence’s openpi provides the  $\pi_0$  [7] VLA model pre-trained on over 10,000 hours of real robot demonstrations, with state-of-the-art results on ALOHA bimanual benchmarks [67]. OpenVLA [27] (7B parameters, 970,000 real-world demonstrations from Open X-Embodiment [43]; MIT license) and Diffusion Policy [15] complete the general-purpose manipulation toolkit. All are general-purpose; none include healthcare-specific training data, safety validation, or clinical task benchmarks.

**Simulation environments.** Genesis (28,254 stars; Apache 2.0) provides unified differentiable physics simulation; Isaac Lab (6,544 stars; BSD-3-Clause) offers GPU-accelerated reinforcement learning with native Unitree G1/H1 support. Neither provides medical environment models, clinical workflow scenarios, or validated patient-contact physics.

Table 2. Open-Source Software Ecosystem for Humanoid and Embodied-AI Robotics (March 2026). Stars are approximate GitHub counts at time of writing. Medical Relevance: **H** = high (direct hardware/algorithm support for clinical candidate platforms), **M** = medium (transferable general capability), **L** = low (infrastructure/simulation only). No repository provides clinical-specific safety validation or medical workflow integration.

| Repository       | Org / Author          | Stars | Purpose                           | License    | Notes                                  | Med. Rel. |
|------------------|-----------------------|-------|-----------------------------------|------------|----------------------------------------|-----------|
| lerobot          | HuggingFace           | ~22k  | VLA/imitation learning pipeline   | Apache 2.0 | Supports Unitree G1; no clinical data  | H         |
| openpi           | Physical Intelligence | ~10k  | $\pi_0$ VLA foundation model      | Apache 2.0 | 10k+ hrs robot demos; bimanual ALOHA   | H         |
| openvla          | Stanford/Berkeley/TRI | ~5.4k | 7B-param open VLA model           | MIT        | 970k demos; generalist manipulation    | H         |
| act              | tonyzhaozh            | ~5k   | Action Chunking Transformers      | MIT        | Original ALOHA bimanual policy code    | H         |
| diffusion_policy | Columbia/MIT          | ~4.5k | Diffusion-based robot policy      | MIT        | State-of-art imitation learning        | M         |
| Genesis          | Genesis-Embodied-AI   | ~28k  | Differentiable physics simulation | Apache 2.0 | No medical environment models          | M         |
| IsaacLab         | isaac-sim (NVIDIA)    | ~6.5k | GPU-accelerated RL + sim          | BSD-3      | G1/H1 native support; no clinical env. | M         |
| unitree_rl_gym   | Unitree Robotics      | ~3k   | RL locomotion for Unitree robots  | BSD-3      | G1/H1/B2 locomotion training           | H         |
| xr_teleoperate   | Unitree Robotics      | ~1.3k | Full-body XR teleoperation        | Apache 2.0 | Used in Atar et al. 2025 study         | H         |
| unitree_sdk2     | Unitree Robotics      | ~0.9k | Hardware SDK for G1/H1/B2         | Apache 2.0 | Low-level hardware interface           | H         |
| ManiSkill        | HKUST                 | ~1.8k | Embodied manipulation benchmark   | Apache 2.0 | No clinical task benchmarks            | M         |
| RoboManip        | isri-robot            | ~0.3k | Bimanual manipulation dataset     | MIT        | Tabletop tasks only                    | M         |
| Open Duck Mini   | apirrone              | ~2k   | Miniature bipedal robot (open HW) | MIT        | Educational; not clinical scale        | L         |

**Unitree hardware ecosystem.** The Unitree GitHub organization maintains nine publicly accessible repositories including `unitree_rl_gym` (3,022 stars), `unitree_sdk2` (932 stars), and `xr_teleoperate` (1,309 stars) for full-body XR teleoperation—the software component used in Atar et al. [4].

**The clinical infrastructure gap.** Four categories of infrastructure required for clinical deployment are absent from the open-source ecosystem: (1) a validated safety layer enforcing patient-contact force limits at runtime; (2) clinical workflow integration connecting humanoid actions to hospital EHR and nursing systems; (3) an open clinical demonstration dataset (every major VLA training corpus consists of household and tabletop tasks); and (4) a validated medical simulation environment with clinical instrument physics and patient anatomy proxies. The transition from general-purpose manipulation to clinically safe, task-verified medical assistance is primarily a software and validation challenge that the open-source community has not yet begun to address.

Table 2 summarizes the most prominent open-source repositories in this ecosystem.

## 5 Medical Applications and Research

The 14 papers identified in the systematic search (Section 2) constitute a thin but internally consistent evidence base. Three cross-cutting patterns define the literature: an almost exclusive reliance on a single commercial platform (Unitree G1), concentration of study production in two research clusters (UC San Diego Yip Laboratory; Johns Hopkins University), and a uniform methodological limitation—no study involves patient subjects in primary contact with a full-size bipedal humanoid. The analysis below leads with cross-study findings within each domain and ends with a critical assessment of study quality and generalizability.

### 5.1 Clinical Procedures and Surgical Assistance

**Cross-study finding.** Teleoperation—not autonomous task execution—is the viable near-term architecture for humanoid clinical procedures. Three studies converge on this conclusion, all using the Unitree G1 as the physical proxy

under remote human cognitive and motor control [4, 16, 30]. Task success across seven clinical procedures reached approximately 70% with non-clinician teleoperators [4]; inverse-kinematics remapping enabled laparoscopic instrument manipulation with enforced remote-center-of-motion constraint [30]; and cadaveric endoscopic surgery was performed with a humanoid as first assistant for the first time on record [16]. Across these three studies, the robot’s contribution is physical presence and dexterity; the clinical judgment and real-time adaptation are supplied by the remote operator. Tasks requiring high force precision (IV catheter insertion, finger-stick testing) showed the highest failure rates, attributable to actuator force resolution and insufficient tactile feedback [4]. Tasks relying on gross positioning (ultrasound probe placement, endoscope holding) showed stronger performance.

**Critical quality assessment.** The procedure evidence is methodologically early-stage. No study involves patient subjects; the earliest data (Atar et al.) uses test fixtures and mannequins. Two of three papers originate from a single laboratory under a single PI (Michael Yip, UCSD), and all three use the Unitree G1. Independent replication on alternative platforms has not occurred. No standardized clinical outcome metric (procedure completion time, complication rate, force safety margin) has been defined or applied across studies. The cadaver study [16] represents the highest pre-clinical evidence level (analogous to bench-to-animal study transition), and identifies three specific engineering targets for next-stage translation: endoscope stability, compliant force control during anatomical contact, and autonomous scope repositioning. TRL assessment: 3–4, uniformly across the three studies.

## 5.2 Elderly and Nursing Care

**Cross-study finding.** The eldercare evidence base is divided between hardware demonstrations with zero patient contact and perception/design studies that do not involve functional humanoid hardware. Benallegue et al. [6] document the only bipedal humanoid designed with nursing as its explicit primary application (RHP Friends), demonstrating transparent autonomous-to-teleoperated handoff at IREX 2023—a clinically relevant architectural principle with no equivalent in commercial platforms. Institutional surveys of 269 US nursing home administrators (Imtiaz and Khan [22]) identify the primary deployment barriers as economic and evidence-based (cost, lack of effectiveness data) rather than technical, a finding that reframes the problem: the bottleneck for eldercare deployment is not engineering capability but regulatory and health-economic evidence. The SPRING project [2] provides the only large-scale deployment data in an eldercare facility (60+ participants over 4 years), but on a wheeled non-bipedal platform (ARI); participants’ preference for upright humanoid form factor for dignified eye-level interaction establishes the user acceptance hypothesis that bipedal systems must replicate with their own evidence.

**Critical quality assessment.** No qualifying paper involves patient contact with a full-size bipedal humanoid in an eldercare context. The user-centered design study (Ghosh et al. [18],  $n = 14$ ) is methodologically sound but uses NAO (58 cm, out of scope for physical care tasks). The bipedal hardware evidence (RHP Friends) is a controlled demonstration, not a clinical study. Generalizability is accordingly low; the perception and institutional survey findings are more transferable than the hardware results. TRL 4 for demonstrated hardware; TRL 4–5 for wheeled platform deployment evidence (scope-constrained).

## 5.3 Rehabilitation

The central engineering gap in humanoid rehabilitation is contact safety: without a validated compliance framework that bounds peak forces during patient-proximate movement, no study involving physical contact between a full-size bipedal humanoid and a patient is ethically or regulatorily feasible. Lu et al. [34] address this directly, demonstrating measurable peak force reduction during quasi-static upper-body contact under the GentleHumanoid spring compliance

framework on the Unitree G1 (TRL 3–4). This is the most technically grounded contribution in the rehabilitation literature, and it sets the engineering precondition for all patient-contact rehabilitation studies that follow.

Two parallel lines of evidence establish why the compliance engineering matters. Nguyen et al. [41] establish a neuroscientific rationale: humanoid robot observation elicits sensorimotor EEG responses comparable to human observation in healthy subjects, supporting an action observation therapy (AOT) basis for stroke rehabilitation. Sobrepera et al. [58] ( $n = 6$ ) document that clinical patients preferred humanoid telepresence over standard video for neuromotor rehabilitation, attributing this to physical embodiment (TRL 4). The convergence of three distinct mechanisms—mirror neuron activation, nonverbal cueing via body posture, and contact-safe compliance—points toward a coherent rehabilitation use case, but the evidence is pre-clinical throughout.

The methodological limitations are sharp. The EEG study enrolled healthy subjects only; replication with patients with motor impairment is required before the AOT rationale translates to a therapeutic claim. The Sobrepera et al. case series is the only study involving actual patients ( $n = 6$ ), and the platform is not a full-size modern bipedal system. No standardized rehabilitation outcome measure (Fugl-Meyer Assessment, 10-meter walk test) has been applied in any humanoid rehabilitation study.

#### 5.4 Mental Health and Social Interaction

The mental health evidence presents an inverse quality-specificity tradeoff that is diagnostic of the field’s current state. The two best-designed studies are randomized: Robinson et al. [52] report a pilot RCT ( $n = 230$ ) in which an autonomous humanoid delivered a brief mindful breathing intervention with positive wellbeing outcomes; Sayis and Gunes [54] find that only the humanoid robot condition (not a voice assistant) produced statistically significant mood improvements and increased self-disclosure ( $n = 42$ ). Both results are credible. Neither, however, specifies the humanoid model used, which prevents extrapolation to full-size bipedal platforms. The platform-specific studies use weaker designs. All studies involve subclinical or university wellness populations; none involves a clinically diagnosed population under treatment. The embodiment advantage is consistent across designs—but demonstrated on unspecified or out-of-scope platforms, in non-clinical settings, with non-patient participants.

The implication for readiness assessment is precise: TRL 4–5 for social and wellbeing applications in non-clinical settings (the RCT evidence supports this claim); TRL 2–3 for bipedal humanoid deployment in clinical mental health contexts (no qualifying evidence exists at this specificity). The LLM personalization framework of Yuan et al. [66]—which models cognitive-emotional states probabilistically for dementia care—is directly applicable to bipedal platforms, though the current implementation uses Pepper. Methodological precedent for automated planning in pediatric clinical settings exists [31]. The next evidential requirement is an RCT with a specified full-size bipedal platform in a clinical population.

#### 5.5 Hospital Logistics

No qualifying papers were found for humanoid-specific hospital logistics tasks performed by bipedal humanoid robots. This absence reflects two structural factors: wheeled autonomous mobile robots already address hospital logistics with proven operational reliability, providing no demonstrated functional advantage for the humanoid form factor in this task category; and academic research attention in 2022–2026 was directed toward the more scientifically novel clinical procedure and HRI domains. TRL for bipedal humanoid hospital logistics: 2 (technology concept formulated; no demonstrated examples in relevant contexts).

Table 3. Evidence Summary by Clinical Domain (as of March 2026). TRL per ISO 16290:2013: 1 = basic principles; 2 = concept formulated; 3 = experimental PoC; 4 = lab validated; 5 = relevant-environment validated; 7 = operational prototype; 9 = proven. \*ARI/SPRING: wheeled platform, not bipedal; scope caveat applies.

| Domain              | Key Papers                                                                                    | Platform             | Study Type                                | N             | TRL     |
|---------------------|-----------------------------------------------------------------------------------------------|----------------------|-------------------------------------------|---------------|---------|
| Clinical procedures | Atar et al. [4], Liang et al. [30], Cho et al. [16]                                           | Unitree G1           | Lab; user study; cadaver                  | 0 patients    | 3–4     |
| Elderly / nursing   | Benallegue et al. [6], Alameda-Pineda et al. [2]                                              | RHP Friends; ARI*    | Exhibition demo; field deploy.            | 0 (demo); 60+ | 4; 4–5* |
| Rehabilitation      | Nguyen et al. [41], So-brepera et al. [58], Lu et al. [34]                                    | Humanoid; Unitree G1 | SRAT; EEG pilot; case series; engineering | 0 / 6 / 0     | 3–4     |
| Mental health / HRI | Robinson et al. [52], Sayis and Gunes [54]                                                    | Unspecified humanoid | hu-RCT; comparative                       | 230; 42       | 4–5     |
| Hospital logistics  | <i>No qualifying papers identified; wheeled autonomous systems dominate this role (TRL 2)</i> |                      |                                           |               | 2       |

## 5.6 Cross-Domain Synthesis

Across the 14 qualifying papers, three structural patterns constrain the generalizability of the current evidence base.

**Platform monoculture.** The Unitree G1 is the experimental platform in 4 of 14 papers (all three clinical procedure studies and the compliance rehabilitation study). No other commercial platform appears in more than one qualifying study. This monoculture is not incidental: as demonstrated in Section 4.4, the Unitree G1’s open SDK is the primary research access point for the commercial humanoid tier. Until equivalent SDK access exists for higher-dexterity platforms (Sanctuary Phoenix Gen 8, 1X NEO Beta), cross-platform replication is structurally blocked.

**Geographic and institutional concentration.** Studies originate predominantly from three countries (USA: 8 papers; Japan: 2; UK: 2; South Korea: 1; multinational EU: 1). The UC San Diego Yip Laboratory accounts for two of the three clinical procedure papers; Johns Hopkins University contributes the third. No study from a clinical institution in the Global South, Southeast Asia, or continental Europe (excluding the SPRING program) appears in the qualifying set. Healthcare workforce demographics, regulatory environments, and care delivery models differ substantially across these absent geographies; findings from US and Japanese studies may not transfer.

**Methodological gaps.** No qualifying study constitutes a randomized controlled trial with a bipedal humanoid robot in a clinical population. No study uses a validated clinical outcome measure as primary endpoint. No study exceeds a single session or demonstration as its primary evaluation context. No study includes health economic analysis or reports cost-per-outcome data. These gaps collectively define the evidence requirements for any future regulatory submission: a prospective, multi-session, clinically powered study on a full-size bipedal humanoid in patient-contact conditions, with a validated clinical outcome measure and a pre-registered analysis plan.

## 5.7 Section Summary

Table 3 summarizes the evidence base across the five clinical domains reviewed above.

The aggregate picture across all five domains is consistent: the field is at TRL 3–4 for bipedal humanoid clinical tasks. Feasibility has been demonstrated in controlled laboratory and pre-clinical settings, primarily by two research groups on a single platform, but no clinical trial with patient subjects, no regulatory submission, and no institutional deployment exists for any full-size bipedal humanoid robot as of early 2026.

Figure 2 visualizes the distribution of the 14 qualifying papers by domain and year, with bubble size proportional to study design strength.

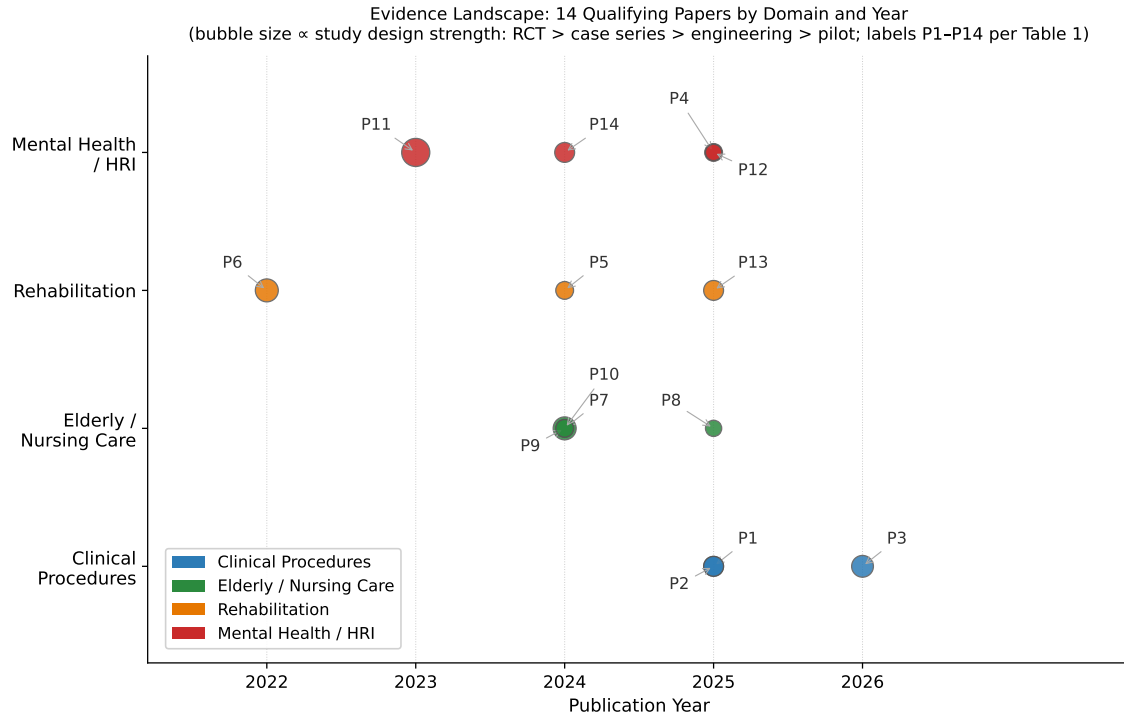


Fig. 2. Evidence distribution: 14 qualifying papers by clinical domain and publication year (bubble size proportional to study design strength: RCT > case series > engineering > pilot). Clinical procedure evidence (blue) is concentrated in 2025–2026 and from a single platform and research cluster. Rehabilitation (orange) and mental health (red) evidence is more temporally distributed but also methodologically limited. No papers qualify for hospital logistics.

## 6 Technical and Translational Challenges

The TRL 3–4 state documented in Section 5 reflects a specific constellation of barriers that are neither uniform nor independently solvable. Some are technical: manipulation precision, contact safety during bipedal locomotion, software validation against clinical performance standards. Some are institutional: regulatory pathway ambiguity, liability framework gaps, actuarial absence. And some are circular: regulatory approval requires clinical evidence; clinical evidence requires IRB-sanctioned trials; IRB approval requires safety assurance; safety assurance requires performance standards that regulatory guidance has not yet established. This section diagnoses each barrier specifically.

### 6.1 Manipulation and Dexterity for Clinical Tools

Atar et al. [4] is the only study to systematically document manipulation performance gaps in a clinical task context. Atar et al. evaluated a Unitree G1 with Inspire Gen4 dexterous hands across seven clinical procedures, achieving approximately 70% aggregate procedural success with non-clinician teleoperators. The paper identifies two specific failure modes: force output insufficiency (the actuator configuration cannot reliably generate the controlled forces required for needle penetration or sustained grip against resistance), and sensor sensitivity limitations (the tactile



feedback loop is insufficient for sub-Newton force resolution, preventing detection of the haptic signatures that guide needle procedures in clinical practice).

These findings generalize across the procedure literature. Liang et al. [30] demonstrate that laparoscopic surgery requires enforcing a remote-center-of-motion (RCM) constraint at the trocar site—an engineering workaround for a capability that commercial humanoid arms do not provide natively. Cho et al. [16] identify three engineering targets for endoscopic clinical translation: endoscope stability in confined anatomical cavities, force compliance during tissue contact, and positioning accuracy for reliable camera view.

Synthesizing across Atar et al., Liang et al., and Cho et al., clinical manipulation requires three capabilities that current commercial humanoid arms do not yet reliably provide: (1) force precision at sub-Newton resolution for needle and catheter procedures [4]; (2) tool-specific kinematic adaptation for laparoscopes, rigid endoscopes, and ultrasound probes; and (3) fingertip-level tactile sensing for grip force discrimination, instrument slip detection, and tissue compliance sensing. Operationally, progress requires force resolution at sub-Newton levels for needle insertion procedures [4], fingertip sensor density exceeding 1,000 sensing elements, and teleoperation round-trip latency at levels consistent with real-time force feedback requirements. Surgical teleoperation research establishes that network latency above 200 ms degrades task completion time and trajectory accuracy in dexterous procedures [35], providing the clinical benchmark for humanoid teleoperation system design. Foundational work on dexterous manipulation documents the multi-dimensional sensor requirements for reliable tool handling [42].

## 6.2 Safe Physical Contact and Bipedal Gait Dynamics

Progress in contact safety has been demonstrated. Lu et al. [34] introduce a spring-based compliance framework for the Unitree G1 that quantifiably reduces peak contact forces during quasi-static tasks including sit-to-stand assistance and upper-body physical contact. This approach builds on the series elastic actuator design [47], in which mechanical spring compliance decouples rapid force transients from the actuated structure. This is the most technically grounded progress toward patient-contact safety in the current literature. Hierarchical whole-body control architectures [55] provide the control-theoretic foundation for multi-priority task execution that safety-critical clinical interaction will require.

The design target for patient-contact force limits is the Power and Force Limiting (PFL) regime of ISO/TS 15066:2016 [24], which provides biomechanically derived thresholds for different body regions: for the face and skull, quasi-static contact force is limited to 65 N and pressure to 110 N/cm<sup>2</sup>; for the trunk and thorax, 140 N and 210 N/cm<sup>2</sup>. The GentleHumanoid framework demonstrates a Unitree G1 approaching these thresholds for quasi-static upper-body contact.

The critical gap, however, is structural. ISO/TS 15066 was designed for fixed-base collaborative robots—cobots whose base is anchored, whose motion is constrained to a defined workspace, and whose approach velocities are programmed. It does not model contact forces arising from bipedal locomotion. A bipedal humanoid navigating a clinical environment introduces four risk scenarios that ISO/TS 15066 does not bound: (1) balance corrections, in which whole-body torque adjustments generate rapid impulse forces that can contact nearby persons; (2) stumble recovery, in which momentum transfer creates swinging limbs and torso rotation at velocities above the standard’s contact speed assumptions; (3) narrow-corridor navigation, in which swinging arms during gait will contact bed rails, IV poles, and patient limbs in spaces with less than one body-width of clearance; and (4) postural transitions, in which a full-size humanoid (35 kg or more) rising to or lowering from a working height near a supine patient generates dynamics that quasi-static force limits do not characterize.

None of the 14 papers reviewed addresses this category of risk, and neither ISO/TS 15066, ISO 13482, nor ISO 10218-1/2 was designed to model or bound gait-induced contact events. This is an open safety engineering problem. Resolution

requires either new standards specific to bipedal humanoid robots in patient-contact environments, or architectural solutions—proximity-triggered gait interrupts, patient-detection zones constraining locomotion near occupied beds, whole-body collision prediction for gait trajectories—that have not yet been developed in the published literature.

### 6.3 Human-Robot Trust and Institutional Adoption

Trust and adoption operate at three distinct layers. The first layer is patient and clinician acceptance. Technology acceptance frameworks established for information systems [63] and their extensions for assistive social robots in elderly care [8, 20] provide the theoretical basis for understanding humanoid adoption barriers. Trust in human-robot interaction is a multidimensional construct influenced by robot performance, human factors, and environmental context [19]. The healthcare-specific question of whether humanoid form factors in clinical settings are perceived as appropriate has been characterized as a strategic and ethical question [46]. Robinson et al. [52] show 53% recruitment uptake for a humanoid wellbeing intervention in a general population (pilot RCT,  $n = 230$ ). Sayis and Gunes [54] show the humanoid condition was the only condition producing measurable mood improvements in a therapeutic writing task ( $n = 42$ ). Both results are bounded: the clinical populations central to humanoid deployment differ substantially from general and educational populations in these studies.

The second layer is co-design and workflow integration. Ghosh et al. [18] demonstrate that iterative co-design with care staff improves system usability—a methodological contribution that does not substitute for clinical evidence of safety and effectiveness.

The third and most institutionally consequential layer is the liability gap. No current healthcare liability framework—US tort law, UK clinical negligence, EU product liability—assigns accountability for harm caused by an autonomous robotic action in a care setting. This legal ambiguity deters adoption regardless of technical capability: hospital risk departments cannot quantify institutional exposure for systems operating in legally uncharted territory. Imtiaz and Khan [22] document this precisely—269 nursing home administrators cite cost, reduction in human contact, and unproven clinical effectiveness as primary barriers.

### 6.4 Regulatory Pathway

The regulatory challenge is not hostility to this technology class; it is a jurisdictional gap that existing frameworks leave precisely where clinical humanoids sit.

ISO 13482:2014 [23] (Safety Requirements for Personal Care Robots) is the most directly applicable safety standard—it governs robots providing physical assistance in human environments. However, §1.4 of ISO 13482 explicitly excludes devices regulated as medical devices. A humanoid robot performing clinical tasks almost certainly meets the medical device definition under FDA 21 CFR §201(h) and EU MDR Article 2(1). This creates a hard gap: the system is likely too clinical for ISO 13482 but no ISO medical robotics standard covers general-purpose bipedal humanoid assistants.

In the United States, the De Novo classification pathway is the only viable regulatory entry route. No 510(k) predicate exists. A De Novo submission requires the applicant to propose Special Controls from scratch: accepted performance standards for fall prevention in patient-care environments, patient contact force limits for dynamic gait scenarios, teleoperation latency bounds for specific procedure types, and AI/ML software validation methodology consistent with FDA’s AI/ML SaMD action plan [61] and December 2024 PCCP guidance [62]. None of these currently exist as accepted standards. This is not procedurally impossible—De Novo has established new device categories for surgical planning software and exoskeletal rehabilitation devices—but it represents a development and regulatory investment no humanoid manufacturer has yet made.

Table 4. Regulatory Framework Comparison for Clinical Humanoid Robots (March 2026). PMS = post-market surveillance; PMCF = post-market clinical follow-up; SaMD = Software as a Medical Device; NB = Notified Body. No framework provides humanoid-specific guidance; all entries represent the authors’ interpretation of existing rules applied to this novel device class.

| Framework             | Jurisdiction | Classification             | Pathway                                | Timeline (est.) | Key Requirements / Gap                                                         |
|-----------------------|--------------|----------------------------|----------------------------------------|-----------------|--------------------------------------------------------------------------------|
| FDA 510(k)            | US           | Class I/II                 | Substantial equivalence to predicate   | 3–12 months     | No predicate exists for clinical humanoid; inapplicable                        |
| FDA De Novo           | US           | Class II (new category)    | Novel device; propose Special Controls | 12–24 months    | Must define fall-prevention, PFL, latency, AI/ML Special Controls from scratch |
| FDA PMA               | US           | Class III                  | Clinical trial evidence required       | 3–7 years       | Overkill for initial classification; may apply for autonomous surgical         |
| EU MDR Class IIa      | EU           | Medium risk                | Notified Body QMS + clinical data      | 18–36 months    | Requires clinical investigation under Art. 62; PMCF post-market                |
| EU MDR Class IIb      | EU           | Higher risk                | Notified Body + full clinical eval.    | 24–48 months    | Autonomous surgical or patient-contact without operator oversight              |
| EU AI Act (High Risk) | EU           | High-risk AI               | Conformity assessment; Art. 6          | From Aug 2026   | Parallel obligation to MDR; dual compliance burden for clinical humanoid       |
| ISO 13482:2014        | Global       | Personal care robot safety | Voluntary standard                     | N/A             | <b>Explicitly excludes medical devices</b> ; no bipedal gait dynamic PFL spec  |
| ISO/TS 15066:2016     | Global       | Collaborative robot HRC    | Voluntary; used in EU MDR              | N/A             | Designed for fixed-base collaborative arms; no dynamic gait scenario coverage  |

The structural problem is a circular dependency: approval requires clinical evidence; human-subject evidence requires IRB approval; IRB approval requires safety assurance; safety assurance requires performance standards; performance standards require regulatory guidance. The productive path through this circle is a staged evidence generation sequence—in vitro feasibility, cadaver study, IRB-approved human-subject feasibility trial—producing the data needed for a De Novo submission. Cho et al. [16] is the current frontier of that sequence.

In the EU, a humanoid clinical assistant would likely be classified as Class IIa under EU MDR Annex VIII, with autonomous surgical assistance escalating to Class IIb or III. EU AI Act Regulation 2024/1689 additionally classifies any AI-driven clinical humanoid as a high-risk AI system. High-risk AI rules apply from August 2026; AI embedded in regulated medical products transitions to August 2027. Any EU clinical humanoid market entry must satisfy both MDR and AI Act simultaneously—a dual compliance burden for which no completed template yet exists.

Table 4 compares the four principal frameworks across the key dimensions that determine market entry strategy.

## 6.5 Software Infrastructure and AI Validation

The leading VLA models—openpi/ $\pi_0$  (10,551 stars), OpenVLA (5,479 stars), and LeRobot (22,179 stars)—represent the highest-capability general-purpose robot learning infrastructure publicly available. For general-purpose humanoid manipulation research, this infrastructure is genuinely strong.

What is absent is the layer between this general infrastructure and clinical deployment. As identified in Section 4.4, the four critical gaps with no current open-source solution are a clinical safety layer, clinical workflow integration, an open clinical demonstration dataset, and a validated medical simulation environment.

The AI validation gap extends to deployment readiness. VLA models achieve high success rates on standard benchmarks, but these share no tasks with the clinical procedure requirements documented in [4]. The distribution shift involves not only visual and geometric differences but also contact physics, tool compliance, and procedural sequencing

Table 5. Technical and Translational Gap Summary (March 2026). Each row identifies a barrier from Section 6, its current state, the threshold required for TRL 5 advancement, which clinical domains it blocks, and the primary evidence or standard defining it. No single gap is insurmountable; they are sequentially dependent in the order listed.

| Capability Gap                                  | Current State                                                                               | Required for TRL 5                                                                                        | Blocking<br>domains | Do-<br>main                              | Key Evidence / Stan-<br>dard      |
|-------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------|-----------------------------------|
| Force precision & tactile sensing               | Insufficient sub-N resolution; no fingertip tactile array on commercial platforms           | Sub-Newton force feedback; continuous fingertip sensing at clinical manipulation precision                | Clinical            | proce-<br>dures, Rehabilitation          | [4]; [35]                         |
| Gait contact safety                             | ISO/TS 15066 quasi-static thresholds approached for upper body only [34]                    | Instrumented characterization of gait-induced force profiles; accepted standard for bipedal PFL           | All                 | physical-<br>contact domains             | ISO/TS 15066 [24]; ISO 13482 [23] |
| Tool-specific kinematics                        | No native RCM; no laparoscope-specific axis on commercial arms                              | RCM enforcement or software analog; tool-path control within trocar constraints                           | Clinical            | proce-<br>dures                          | [30]; [16]                        |
| VLA clinical generalization                     | VLA benchmarks: household manipulation only; no clinical demonstration dataset exists       | Open clinical demonstration dataset; VLA fine-tuned on clinical task protocols with validated performance | Clinical            | pro-<br>cedures, all<br>autonomous roles | [43]; [67]                        |
| Clinical safety standard & regulatory predicate | No accepted standard for bipedal gait in patient-contact environments; no De Novo predicate | ISO scope revision; IRB feasibility study; FDA De Novo with proposed Special Controls                     | All domains         |                                          | [23]; [61]; [62]                  |

requirements fundamentally different from household manipulation. AI in health and medicine more broadly faces similar validation challenges as these systems move toward clinical deployment [50, 60]. No humanoid software stack has addressed IEC 62443 Security Level 2 requirements either—the minimum threshold for networked systems in hospital environments.

Table 5 synthesizes the five principal technical and translational barriers identified in this section.

## 7 Discussion and Future Outlook

The evidence surveyed in Sections 5 and 6 presents a field at an inflection point: capable enough to demonstrate clinical relevance in controlled settings, yet short of every threshold—safety engineering, clinical evidence, and regulatory precedent—required for operational deployment.

### 7.1 Technology Readiness Assessment

The TRL assignments in Table 3 provide the starting point. The aggregate finding—TRL 3–4 across all domains for bipedal humanoid platforms—is not a uniform statement; the TRL ceiling for each domain is governed by a specific domain-particular barrier.

**Clinical procedures (current TRL 3–4).** Atar et al. [4], Liang et al. [30], and Cho et al. [16] constitute the most technically substantive evidence in the review. The cadaveric study of Cho et al. is the current frontier of the evidence sequence: the step immediately preceding an IRB-approved human-subject feasibility trial. Advancing to TRL 5 requires

two things simultaneously: a structured IRB-approved feasibility study in a specific narrow indication (telemedicine-assisted physical examination is the most defensible starting point, because fully teleoperated operation keeps a human in the decision loop at all times), and independent replication on a platform other than the Unitree G1.

**Elderly and nursing care (TRL 4–5 for social robots; TRL 3–4 for bipedal humanoids).** The gap between platform classes is analytically significant. The SPRING project achieved TRL 4–5 through real-world deployment with 60+ older adult users over four years [2]. RHP Friends—the only purpose-built bipedal nursing humanoid—demonstrated patient transfer tasks at IREX 2023 but has not been evaluated in an actual care facility [6]. The path to TRL 5 for bipedal humanoids is not a question of platform capability but of deployment context: a structured pilot in an actual care facility, using SPRING-derived methodology on a bipedal platform, is the specific experiment that would advance this domain.

**Rehabilitation and mental health (TRL 3–4 and TRL 4–5, respectively).** These two domains share a structural problem: the strongest available evidence was generated on the wrong platforms. In rehabilitation, Nguyen et al. [41] establish neuroscientific rationale, Sobrepera et al. [58] establish clinical acceptability in a small case series, and Lu et al. [34] establish the contact-safety engineering foundation—but none of the three used a full-size modern bipedal humanoid with a clinical patient population. In mental health, Robinson et al. [52] ( $n = 230$  RCT) and Sayis and Gunes [54] ( $n = 42$ ) achieve methodologically strong results with consistent embodiment advantage findings—but neither specifies the humanoid model, and all participants are non-clinical wellness populations.

The advance required in both domains is the same experiment in different forms: an IRB-approved human-subject study using a specified full-size bipedal platform with a defined patient population. For rehabilitation, the target is a stroke trial with standardized outcome measures (Fugl-Meyer Assessment, 10-meter walk test). For mental health, the target is a feasibility trial with a clinically diagnosed population assessing validated scales. Both domains are one targeted study away from TRL 5. Both are currently blocked by the absence of that study, not by lack of underlying evidence.

**Hospital logistics (TRL 2).** No peer-reviewed evidence exists for humanoid-specific hospital logistics tasks. Wheeled autonomous mobile robots already address this domain with demonstrated operational reliability. Hospital logistics is not a priority for bipedal humanoid development at this stage.

**Cross-domain finding.** No domain has reached TRL 7 (system prototype demonstrated in operational environment). TRL 5—the threshold of first human-subject validation in a controlled clinical setting—is the realistic five-year target for the three leading domains.

Figure 3 visualizes the current TRL state and TRL 5 target across all five clinical domains.

Figure 4 quantifies the specific capability gaps between the current state-of-the-art and the required thresholds for two leading clinical deployment domains.

## 7.2 Near-Term Deployable Roles

Three roles have the strongest case for TRL 5 achievement within five years.

**Role 1: Telemedicine and telepresence clinical assistant.** The teleoperated mode is the key risk-mitigation design choice: by keeping a human operator in the decision loop at all times, it avoids the regulatory classification and safety validation burdens associated with autonomous patient contact. The technical infrastructure exists: Atar et al. [4] demonstrate teleoperated clinical task execution across seven procedures, and Liang et al. [30] extend this to laparoscopic surgery. The primary application is specialist access extension—enabling a remote specialist to conduct a

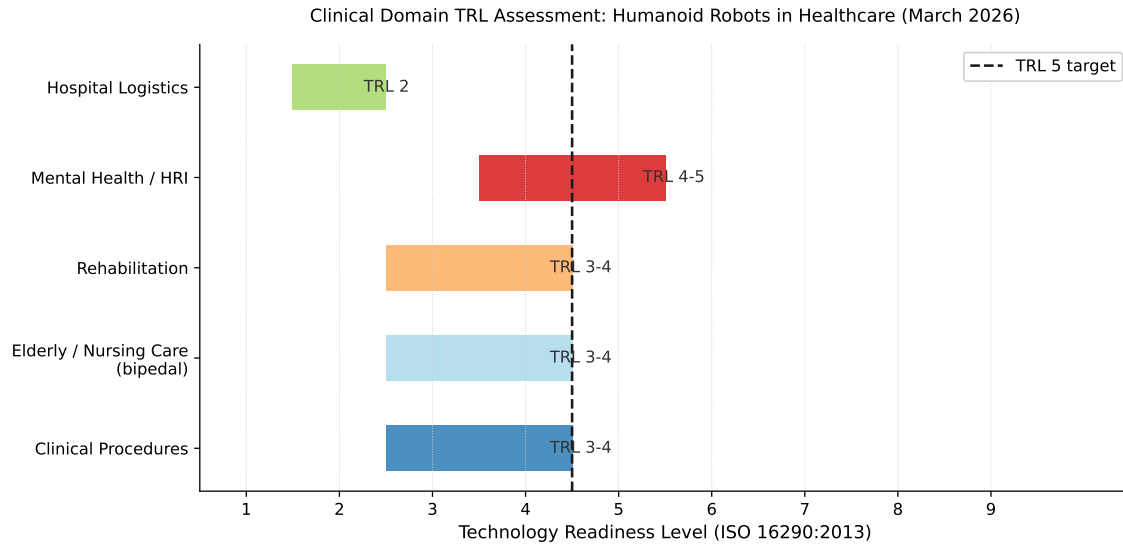


Fig. 3. Technology Readiness Level (TRL) assessment by clinical domain (March 2026). Current TRL range (filled bars) and TRL 5 target (dashed line) per ISO 16290:2013. No domain has reached TRL 7 (operational environment prototype); TRL 5 (relevant-environment validation) is the realistic five-year target for clinical procedures, elderly care (bipedal), and rehabilitation.

physical examination or assist in a procedure in a rural or low-resource setting. Realizing this role at TRL 5 requires a defined physical examination protocol, remote operator training standards, and IRB oversight.

**Role 2: Rehabilitation movement demonstrator.** This role requires no direct patient contact: the humanoid demonstrates movements that the patient observes or follows. The bipedal gait safety gap identified in Section 6.2—the most universally blocking technical barrier—becomes irrelevant to this specific role, because the safety concern is dynamic contact force during locomotion near patients, not movement observation at distance. The neuroscientific rationale is grounded in Nguyen et al.’s mirror neuron evidence [41]; the clinical acceptability is established by Sobrepera et al.’s case series [58]. Advancing requires a registered feasibility trial adapting a NAO or Pepper protocol to a full-size bipedal humanoid demonstrator in stroke rehabilitation.

**Role 3: Structured eldercare companion and physical assistant.** RHP Friends [6] is the only platform designed specifically for this role. The hybrid autonomous/teleoperated architecture—in which routine tasks run autonomously and non-routine tasks are handed to a remote operator—is the correct design for this deployment scenario. The SPRING project [2] provides the deployment methodology and user-acceptance evaluation framework. Realizing this role at TRL 5 requires an actual care facility pilot.

### 7.3 Research Priorities

Four priorities, ordered by their blocking impact on the entire field:

**Priority 1: Clinical safety validation for bipedal gait in patient-contact environments.** The four gait-induced contact risk scenarios documented in Section 6.2 are not bounded by any existing standard. No IRB will approve human-subject studies involving physical patient contact by a full-size bipedal humanoid without instrumented biomechanical

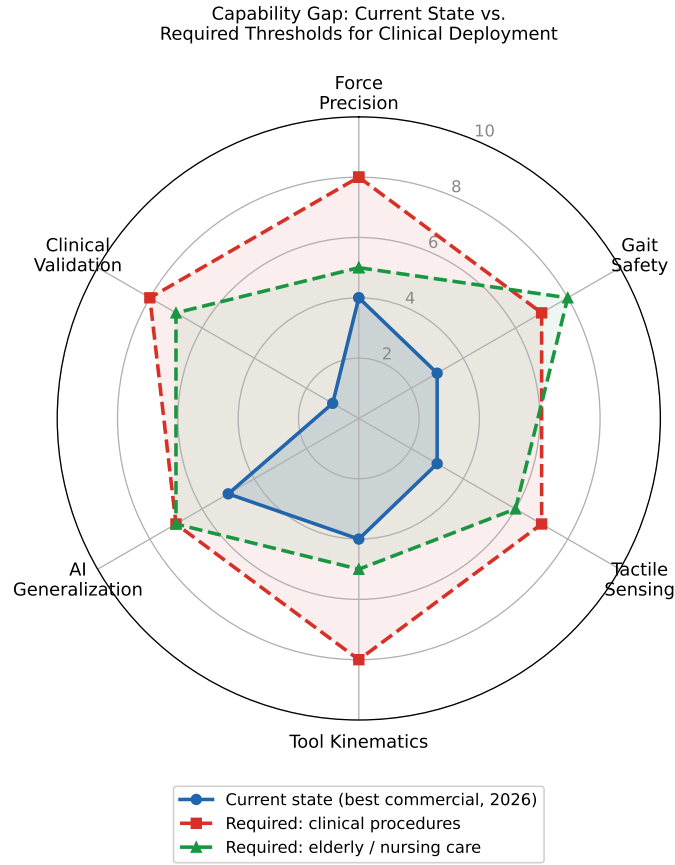


Fig. 4. Capability gap radar chart: current state-of-the-art (blue, 2026) versus required thresholds for clinical procedures (red dashed) and elderly nursing care (green dashed). Scores are on a 0–10 scale representing engineering and evidence maturity. Clinical Validation is the universal bottleneck (current score 1 of 10); Gait Safety is the dominant engineering gap for eldercare; Force Precision and Tool Kinematics are the key gaps for clinical procedure performance.

data on these force profiles and an accepted safety standard or architectural safeguard. This is the single most leverage-intensive investment the field can make: resolving it unblocks physical patient-contact studies in all three leading domains simultaneously.

**Priority 2: Human-subject feasibility studies in leading applications.** The cadaveric endoscopy study of Cho et al. [16] is the current clinical frontier. The specific next step is an IRB-approved, registered feasibility study in a narrow indication with an explicit safety monitor protocol. UC San Diego (Yip Laboratory) and Johns Hopkins are the natural sites.

**Priority 3: Open clinical demonstration dataset for VLA training.** No public dataset of humanoid robot demonstrations in clinical or clinical-simulated environments exists. Creating an open clinical demonstration dataset—even using simulators and clinical task protocols with mannequins—would unlock foundation model fine-tuning for clinical applications and enable the broader research community to advance clinical policy learning.



Table 6. Humanoid Clinical Deployment Readiness (HCDR) framework applied to near-term deployable roles (March 2026). Binding constraint is the axis whose gap most delays deployment. TRL = Technology Readiness Level (ISO 16290:2013).

| Role                            | Autonomy                  | Contact                | Reg. Complex-<br>ity | Complex-<br>ity | Binding<br>constraint                |
|---------------------------------|---------------------------|------------------------|----------------------|-----------------|--------------------------------------|
| Telemedicine assistant          | Teleoperated              | Indirect (tools)       | Class De Novo        | II /            | Safety standard: bipedal gait        |
| Rehabilitation demonstrator     | Supervised                | None                   | Class I–II           |                 | Evidence: patient trial              |
| Eldercare companion / assist    | Supervised-to-conditional | Direct                 | Class IIb–III        |                 | Both: gait safety + contact standard |
| Laparoscopic surgical assistant | Teleoperated              | Indirect (instruments) | Class III            |                 | RCM precision + regulatory precedent |

**Priority 4: Platform diversification for evidence generalizability.** The concentration of clinical procedure evidence on the Unitree G1 (Atar et al., Liang et al., Cho et al.) is a scientific limitation. Replication of clinical task evaluations on RHP Friends, TALOS, or other platforms would establish whether current findings are platform-specific or generalizable to the bipedal humanoid class.

#### 7.4 Humanoid Clinical Deployment Readiness Framework

The TRL scale captures technology maturity along a single axis. Clinical deployment decisions require reasoning along at least three independent axes simultaneously: *autonomy level*, *patient-contact intensity*, and *regulatory complexity*. We introduce the Humanoid Clinical Deployment Readiness (HCDR) framework to make this three-dimensional space explicit. The framework is not a scoring system; it is a classification tool for locating specific role-platform combinations in the deployment readiness space and identifying which axis is the binding constraint.

**Axis 1: Autonomy level.** Roles range from fully teleoperated (human operator controls every action in real time) through supervised autonomy (the robot executes programmed sub-tasks under human oversight and with human abort authority) to conditional autonomy (robot operates autonomously under defined operational conditions with human escalation) and full autonomy (unsupervised operation). Current evidence supports only fully teleoperated roles. No qualifying study demonstrates supervised autonomy in a patient-contact task.

**Axis 2: Patient-contact intensity.** Contact ranges from no direct contact (observation, companionship, logistics) through indirect contact (physical interaction via tools or instruments, patient not in direct force chain) to direct physical contact (hands on patient, transfer assistance, examination). The binding regulatory and safety constraints scale sharply with contact intensity: indirect tool contact requires force control validation; direct physical contact requires validated safety standards that do not yet exist for bipedal platforms.

**Axis 3: Regulatory complexity.** Complexity is determined by device classification (Class I–III under FDA; Class I–III under EU MDR), presence of AI/ML as a decision-making component, and whether the application involves therapeutic intent (triggering clinical investigation requirements) or supportive logistics (lower classification pathway). Full autonomy with direct patient contact and therapeutic intent is the highest-complexity combination; teleoperation for physical examination assistance with physician oversight is among the lowest.

Table 6 maps the three near-term deployable roles against the HCDR framework and identifies the binding constraint on each path.

The framework reveals that the binding constraint is not uniform across roles. For the telemedicine assistant, the primary block is a missing safety standard for bipedal gait in patient-proximate environments—an engineering and standards-body problem, not a clinical evidence problem. For the rehabilitation demonstrator, the block is clinical evidence: the safety case is straightforward (no direct contact), but no powered clinical trial exists. For eldercare physical assistance, both gait safety and direct-contact force standards are missing simultaneously. The HCDDR matrix operationalizes this distinction: investment in gait safety standards unblocks the telemedicine and eldercare roles; investment in a rehabilitation trial can proceed independently without waiting for standards resolution.

## 7.5 Practitioner Guidance

The evidence synthesized in this review has different implications for different practitioner audiences. This subsection articulates those implications explicitly.

**For clinical engineers and hospital technology leadership.** The commercial humanoid platforms entering the market in 2024–2026 are designed for industrial labor, not healthcare. Procurement decisions made in 2026–2027 should distinguish between research-access platforms (Unitree G1: open SDK, lowest cost, only platform with published clinical task evidence, but physically limited) and future clinical platforms (RHP Friends: purpose-built but academic/prototype-only; Sanctuary Phoenix Gen 8: highest haptic sensing specification but closed ecosystem). No commercial platform is ready for unsupervised clinical use in any application. Any hospital-based evaluation program should structure the initial phase as an IRB- approved research protocol, not an operational pilot, to ensure safety monitoring, data collection, and liability clarity. Partnering with an academic medical center (not a robotics company) to co-design the evaluation protocol is the appropriate model.

**For robotics engineers entering the healthcare domain.** The most impactful contribution from the engineering community is not yet another manipulation algorithm or VLA architecture—those are not the bottleneck. Four specific infrastructure gaps documented in Section 4.4 represent engineering problems with no current open-source solutions: (1) a runtime safety layer enforcing ISO/TS 15066 contact force limits for bipedal dynamic scenarios; (2) clinical workflow integration middleware connecting humanoid action APIs to HL7 FHIR and hospital EHR systems; (3) a public clinical demonstration dataset for VLA fine-tuning on medical procedures; and (4) validated medical simulation environments with patient anatomy proxies and clinical instrument physics. Any of these would materially advance the field more than platform capability improvements at current TRL levels.

**For clinical researchers and trial designers.** Three study designs are currently feasible and would individually advance the field: (1) A registered, IRB-approved feasibility study of teleoperated humanoid physical examination assistance in a telemedicine-enabled rural clinic or outpatient setting—the lowest-complexity entry into patient-contact evidence. (2) A pilot RCT of humanoid-delivered action observation therapy for stroke rehabilitation, adapting the Sobrepera et al. protocol to a full-size bipedal platform and powered for a functional outcome measure (Fugl-Meyer Assessment). (3) A prospective multi-session care facility pilot adapting SPRING methodology to RHP Friends or an equivalent purpose-built platform, measuring care quality and staff burden outcomes. All three designs require a pre-specified safety monitoring protocol and an a priori definition of the study-stopping criteria for unexpected adverse events.

## 7.6 Regulatory Roadmap

Figure 5 illustrates the parallel regulatory tracks and their key decision points for both jurisdictions.

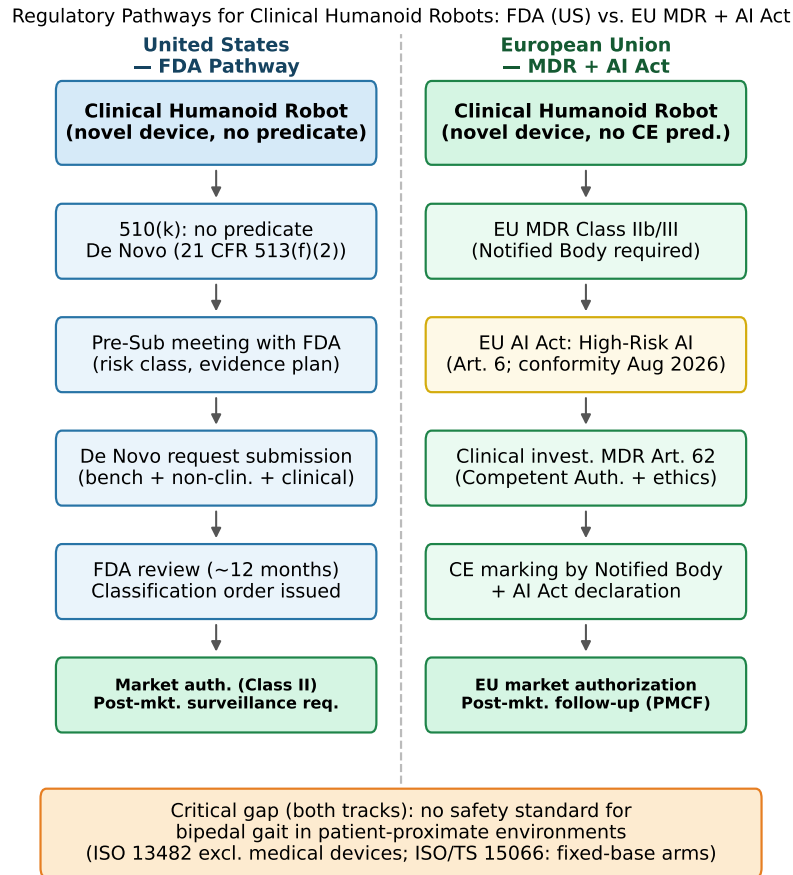


Fig. 5. Regulatory pathways for clinical humanoid robots: FDA De Novo (US, left) and EU MDR plus EU AI Act (EU, right). Both tracks require the same foundational evidence—safety standards for bipedal gait in patient-proximate environments—that does not yet exist (bottom banner). FDA De Novo is identified as the most plausible first-clearance pathway; EU MDR plus AI Act dual compliance represents the EU parallel track.

The path to the first regulatory clearance requires three sequential steps.

**Step 1: Performance standards definition.** A joint effort among humanoid manufacturers, academic medical centers (UCSD, JHU), and FDA’s Center for Devices and Radiological Health (CDRH) is required to define the Special Controls that any De Novo submission must meet: fall prevention performance requirements, patient contact force limits for dynamic bipedal scenarios, teleoperation latency bounds, and AI/ML software validation methodology consistent with FDA’s Good Machine Learning Practice principles and the December 2024 PCCP guidance.

**Step 2: IRB feasibility study generating clinical evidence.** A registered, IRB-approved feasibility study in a specific narrow indication generates the clinical evidence required for a De Novo submission. This study must be

designed from the outset with regulatory submission in mind: pre-specified safety endpoints and a defined safety monitor protocol.

**Step 3: De Novo submission establishing a regulatory predicate.** The De Novo application proposes a new product code for bipedal humanoid clinical assistants with Special Controls from Steps 1 and 2. First clearance establishes a predicate device, enabling subsequent 510(k) applications.

For the EU, the parallel path requires: an ISO/TC 299 scope revision of ISO 13482 to include clinical humanoids; EU MDR Class IIa conformity assessment; and EU AI Act compliance under high-risk AI obligations (effective August 2026 for new AI systems; August 2027 for AI embedded in regulated medical products). Regulatory precedents for managing layered framework compliance exist in the surgical robotics domain.

## 7.7 Ethical Considerations

**Dignity of care.** Physical care interactions require that the patient feels cared for, not processed. The SPRING project finding that participants cited eye-level conversational positioning as important for dignified interaction [2] provides direct empirical evidence that form factor affects perceived dignity of care. The uncanny valley phenomenon—the negative affective response elicited by human-like but not fully human entities [39]—is particularly relevant for humanoid design in care settings, where distress rather than curiosity is the cost of a poor perceptual response. Care robotics designers must incorporate patient dignity explicitly as a design criterion alongside technical performance metrics.

**Liability and informed consent.** No current healthcare liability framework assigns accountability for harm caused by an autonomous or semi-autonomous robotic action in a care setting. The ethical dimensions of robot care for elderly individuals—including dignity, autonomy, and emotional manipulation risks—require explicit frameworks before deployment [56]. Before any human-subject deployment, consent processes for humanoid-assisted care must be defined. These require deliberate engagement between legal, ethical, and clinical stakeholders before the first human-subject study is launched.

**Data privacy.** A humanoid operating in a patient room integrates cameras, microphones, depth sensors, physiological monitors, and, in future deployments, integration with EHR systems, into a single mobile platform in one of the most privacy-sensitive environments imaginable. No data governance framework specific to clinical humanoid sensor data currently exists.

**Workforce framing.** The evidence does not support a displacement narrative. No clinical role exists that a humanoid currently performs reliably enough to replace a trained nurse or clinician. The evidence-supported framing is task offloading: routine, physically demanding, or cognitively repetitive tasks that do not require clinical judgment could be offloaded, freeing clinicians for judgment-requiring activities. The administrator survey of Imtiaz and Khan [22] shows that the perceived risk of reduced human contact is a primary institutional barrier; communicating the task-offloading framing accurately to institutional decision-makers is as important as resolving the technical barriers.

## 8 Conclusion

As of March 2026, no bipedal humanoid robot holds FDA clearance or CE marking for any clinical application. No registered Phase I, II, or III clinical trial has enrolled patients for a study involving a full-size bipedal humanoid platform. The systematic search underlying this review identified 14 qualifying papers across 2022–2026—a literature sparse enough to read in its entirety, concentrated on a single commercial platform (Unitree G1) and, for the clinically most advanced domain, on a single research cluster (UCSD Yip Laboratory and Johns Hopkins). That concentration is not

evidence of a field in failure; it is evidence of a field in its earliest phase of organized research. The infrastructure that would enable distributed clinical investigation—accepted safety standards for bipedal patient-contact environments, IRB-validated study protocols, and regulatory pathways with established Special Controls—does not yet exist, and its absence explains the concentration far better than any deficit of platform capability or clinical motivation.

The contribution of this review is to document that gap with specificity sufficient to act on it. Prior reviews of healthcare robotics, including the closest comparative work in the literature [17], either predate the LLM/VLA inflection that is the enabling technical development of the current period, or treat humanoid and non-humanoid platforms interchangeably, obscuring the platform-specific barriers and advantages that define the clinical deployment problem. The synthesis here is the first to integrate the full 2022–2026 humanoid-specific literature with a commercial and academic platform capability assessment, a regulatory pathway analysis, and a per-domain TRL readiness evaluation grounded in clinical deployment criteria.

The field is not uniform in its readiness. Three roles have a credible path to first human-subject validation—TRL 5—within five years: a telemedicine and telepresence clinical assistant operating in fully teleoperated mode, which minimizes autonomous patient-contact risk and maps directly onto the established regulatory logic of physician-controlled surgical robotics; a rehabilitation movement demonstrator that exploits the mirror neuron mechanism established by Nguyen et al. [41] and the clinical acceptability demonstrated by Sobrepera et al. [58] without requiring any patient contact, rendering the bipedal gait safety gap irrelevant to this role; and a structured eldercare companion and physical assistant following the hybrid autonomous/teleoperated architecture demonstrated by RHP Friends [6] and the deployment methodology established by the SPRING project [2]. These are not speculative roles. They are identified by triangulating the strongest evidence in the literature, the most tractable technical barriers, and the lowest regulatory risk profile available to each application.

The HCDR framework introduced in Section 7.4 makes this role-differentiated picture precise. By classifying roles along three axes—autonomy level, patient-contact intensity, and regulatory complexity—the framework reveals that the binding constraint is not the same for every application. For the telemedicine assistant role, the block is a missing gait safety standard; for the rehabilitation demonstrator, it is clinical evidence; for eldercare physical assistance, both are required simultaneously. This decomposition has a practical consequence: investment in gait safety standards and investment in a rehabilitation feasibility trial are independent activities that can proceed in parallel. The field does not face a single bottleneck; it faces several distinct ones, each amenable to a focused, bounded intervention.

Two gaps are more consequential than the others identified in Section 6. The first is the absence of any accepted performance standard for bipedal humanoid robots in patient-proximate environments—specifically for the four categories of gait-induced contact events that ISO/TS 15066 was not designed to address. This gap does not require a breakthrough to close; it requires a biomechanical characterization study and the standards development process. But no IRB will approve physical patient-contact studies without it, and no regulatory agency can specify Special Controls without it. It is the single most globally blocking gap in the field. The second is the absence of registered human-subject feasibility studies. Cho et al. [16] advance the state of the art by completing a cadaveric surgical study under a defined IRB protocol—the step immediately preceding first-in-human research. UC San Diego and Johns Hopkins, which together hold all three papers in the clinical procedures domain, are the natural sites for this next step. There is no institutional reason, beyond the gait safety gap, why this study cannot begin within the current funding cycle.

The question confronting the field is not whether humanoid robots can play a useful role in healthcare. The evidence in this review establishes that they can—not at the level of clinical deployment, but at the level of demonstrated laboratory capability in procedures and care tasks that have genuine clinical value. The question is whether the research

community, device manufacturers, clinical institutions, and regulatory agencies will commit to the specific, sequential, methodical process required to convert that laboratory evidence into clinical evidence, and clinical evidence into regulatory precedent. That process—safety characterization, IRB feasibility study, De Novo submission, and predicate device establishment—is well understood in medical device development. Applying it to a genuinely novel platform class, with novel form-factor dynamics and novel AI system integration, requires institutional investment and coordination that has not yet materialized at scale. The evidence base is ready for the next step. The infrastructure to take it is the work of the next five years.

## Acknowledgments

No external funding was received for this work.

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